

7a	energy from 1 nucleus = $(1.77 \times 10^{13}) / (6.02 \times 10^{23})$ $(= 2.94 \times 10^{-11} \text{ J})$ Energy released = Binding energy of products – Binding energy of reactants binding energy of Z = $[(1.25 + 1.81) \times 10^{-10}] - 2.94 \times 10^{-11}$ □ $(= 2.77 \times 10^{-10} \text{ J})$ nucleon number of Z = $93 + 139 + 2 - 1$ $(= 233)$ Binding energy per nucleon of Z = $(2.77 \times 10^{-10}) / (233 \times 1.60 \times 10^{-13}) = 7.43 \text{ MeV (3 s.f.)}$	C1 C1 C1 A1
7bi	$N_0 = 0.874 / (238 \times 1.66 \times 10^{-27}) = 2.212 \times 10^{24}$ $= 2.21 \times 10^{24}$	A1

7bii	$A = \lambda N = \frac{\ln 2}{t_{\frac{1}{2}}} N$ $= \frac{\ln 2}{87.7 \times 365 \times 24 \times 3600} \times 2.21 \times 10^{24}$ $= 5.54 \times 10^{14} \text{ Bq}$	C1 A1
7biii	$A = A_0 e^{-\lambda t}$ $\ln 0.653 = -\left(\frac{\ln 2}{87.7}\right)t$ $t = 53.9 \text{ years}$	C1 A1
7biv	half-life shorter, will not provide power for long enough / require frequent replacement of probe	B1

8ai	Horizontal component of tension / spring force provides centripetal force	B1
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